

SUDHANSHU PAL

SENIOR BACKEND ENGINEER

Java · Spring Boot · Microservices · Apache Kafka · Kubernetes · AWS · Spring AI
+91 70078 23554 · sudhanshupal03616@gmail.com · Noida, India
sudhanshu-portfolio-nu.vercel.app · linkedin.com/in/sudhanshupal7408 · github.com/Sudhanshup7408

PROFESSIONAL SUMMARY

Senior Backend Engineer with 4+ years designing and scaling secure, high-throughput Java and Spring Boot services across fintech and government platforms. Builds event-driven microservices on Apache Kafka and Redis at 1M+ daily transaction scale, with REST API design, OAuth 2.0/RBAC security, and PostgreSQL/MongoDB persistence. Replaced a licensed commercial document library with a PDF and media engine written from scratch against the specification, saving ~\$15K per year. Hands-on across Docker, Kubernetes, Jenkins CI/CD, and Prometheus/Grafana/ELK observability, plus production Spring AI and RAG delivery. Published IEEE conference author.

TECHNICAL SKILLS

Languages: Java (8/11/17), SQL, Bash/Shell, Core Java, Multithreading, Concurrency, OOP, SOLID
Frameworks: Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate/JPA, REST APIs, gRPC
Architecture: Microservices, Event-Driven Architecture, Distributed Systems, API Gateway, CQRS, Saga
Messaging: Apache Kafka, RabbitMQ, WebSocket, Server-Sent Events (SSE), Webhooks, Async Processing
Databases: PostgreSQL, MySQL, MongoDB, Redis, Schema Design, Indexing, Query Optimization
Documents & Media: PDF specification, iText, Apache PDFBox, FFmpeg, PDF encryption, Annotations, Magic-number file type detection, Image/Video transcoding, DOCX to PDF
DevOps & Cloud: Docker, Kubernetes, Jenkins, CI/CD, AWS (EC2, S3, IAM), Nginx, Linux, Maven, Git
Observability: Prometheus, Grafana, ELK Stack (Elasticsearch, Logstash, Kibana), Alerting, RCA
Security: OAuth 2.0, JWT, RBAC, SSO, OWASP Top 10, SonarQube, Checkmarx, SAST, TLS/SSL
AI / LLM: Spring AI, LLM APIs, RAG, Vector Databases (pgvector), Embeddings, Semantic Search
Testing: JUnit 5, Mockito, TDD, JaCoCo, JMeter, Integration Testing, Code Coverage
Practices: Agile/Scrum, System Design, Code Review, Production Support, Incident Management, Client Handling

PROFESSIONAL EXPERIENCE

PeLocal Fintech Private Limited <i>Senior Backend Developer</i>	Feb 2026 – Present Noida, India
• Architected Spring Boot microservices and REST APIs powering Tourism, Police, and Healthcare portals. • Built a multi-tenant WhatsApp bot on Meta Cloud API routing three departments through separate flows. • Hardened webhook ingestion with signature verification, message dedup, and Kafka-backed processing. • Delivered a RAG pipeline over pgvector embeddings and semantic search, cutting support queries ~40%. • Streamed token-by-token LLM output to clients over WebSocket and SSE for real-time conversations. • Designed polyglot persistence with PostgreSQL for transactions and MongoDB for reporting aggregates. • Implemented OAuth 2.0 and four-tier RBAC in Spring Security, securing REST APIs with JWT and SAST. • Built the Agent-Supervisor CRM backend with SLA-tracked ticketing at 500+ tickets/day and KPI reporting. • Containerized services with Docker and automated Jenkins CI/CD releases to Kubernetes. • Instrumented Prometheus metrics, Grafana dashboards, and centralized ELK logging across all services.	
Paytm <i>Software Developer</i>	Sep 2025 – Feb 2026 Noida, India
• Migrated three legacy partner flows to event-driven microservices on Apache Kafka and RabbitMQ. • Lifted transaction success rates 10-15% across the migrated partner integration flows. • Debugged critical Kafka pipeline and Redis cache failures across four incidents with zero data loss. • Cut MTTR 30% by triaging live production incidents through Kibana log analysis and monitoring. • Integrated four insurance provider APIs into production with zero customer-facing disruption.	

Newgen Software Technologies (NSTL)

Senior Software Design Engineer

Jun 2023 – Sep 2025

Noida, India

- Rebuilt the document processing stack from scratch against the PDF specification with no third-party library, removing the licensed iText dependency and saving ~\$15K per year in licence cost.
- Implemented page-level PDF manipulation: create, insert, delete, remove, shuffle, split, and merge.
- Built PDF security in-house, encrypting and decrypting documents with owner and user passwords and per-operation permission flags.
- Developed the full annotation layer: line, circle, rectangle, free text, link annotations, and text and image watermarks.
- Detected true file type from magic-number headers rather than file extensions, rejecting mislabelled uploads before parsing.
- Engineered bidirectional media conversion, image sequences to video and video to extracted frames, plus DOCX to PDF conversion APIs.
- Developed a Spring Boot video KYC service with OpenVidu and FFmpeg on AWS EC2 for compliance capture.
- Owned production support end to end, triaging live tickets and working directly with clients from first report through root cause to release.
- Improved service throughput 35% through multithreading optimization, JMeter-validated at 500 concurrent users.
- Resolved 20+ production issues, reached 90%+ unit coverage with JUnit, Mockito, and JaCoCo, and cleared all SonarQube and Checkmarx findings before release.

National Payments Corporation of India (NPCI)

Graduate Engineer Trainee

Jun 2022 – Feb 2023

Hyderabad, India

- Designed a microservice architecture supporting 1M+ daily transactions with Kafka event processing.
- Applied Redis caching and circuit-breaker patterns to sustain high availability under peak load.
- Built Core Java and Hibernate/JPA services for NACH, processing 1M+ daily financial transactions.
- Automated license notification workflows that cut manual effort 60%.
- Delivered BBPS Biller Dashboard 2.0 REST APIs with Swagger/OpenAPI docs for on-time Day-1 go-live.

KEY PROJECTS

Document Processing Engine · Newgen. PDF and media engine written from scratch against the specification with no third-party library. Page operations, encryption, annotations, magic-number type detection, and image/video conversion. Replaced iText and removed ~\$15K per year of licence cost.

Citizen Services Platform · PeLocal. Three government departments on one multi-tenant Spring Boot backend with tenant-scoped routing and isolated data boundaries.

Conversational AI Backend · PeLocal. Spring AI orchestration, pgvector retrieval, and WebSocket/SSE token streaming; cut inbound support queries ~40%.

Agent & Ticketing CRM · PeLocal. Four-role access model, live chat, and SLA automation at 500+ tickets/day with supervisor KPI reporting.

Video KYC Service · Newgen. OpenVidu and FFmpeg media pipeline on AWS EC2 for compliance capture and regulatory retention.

NACH & BBPS Payment Rails · NPCI. Core Java and Hibernate services sustaining 1M+ daily transactions on national payment infrastructure.

PUBLICATION

Effectiveness of Machine Learning Algorithms in Forecasting AQI

D. Pramanik, E. Jain, S. Pal · IEEE conference, Tashkent, Uzbekistan · November 2021

Compared regression and tree-based models for air quality index forecasting; Decision Tree performed best on small datasets and Random Forest on larger ones. ieeexplore.ieee.org/document/9673385

EDUCATION

B.Tech, Computer Science & Engineering

DIT University

Jun 2018 – May 2022

Dehradun, India

CERTIFICATIONS & AWARDS

Pinnacle Performance Award · Newgen Software (NSTL), for outstanding delivery and zero-defect production releases.

Java Basic to Advance · IIT Kanpur

Introduction to Java · Microsoft MTA